

2024



INSTABASE

Building Generative AI Solution AI Hub Using AWS with Instabase

Learn how Instabase in the software and internet industry built AI Hub, a generative AI solution based on large language models, using AWS.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

20% improvement

in performance using Amazon EC2 M6i

30% reduced costs

70% reduced

development costs using Amazon EC2

Reliable and highly available

solutions achieved



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Improved speed

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Overview

Applied artificial intelligence (AI) company [Instabase Inc.](#) is on the leading edge of generative AI (GenAI) and serves multiple top 10 global banks and other large enterprises such as Axa and Sonic Automotive. Because training machine learning (ML) models is a complex process that often requires hundreds of samples and weeks of developer time, Instabase wanted to make a product that was simple for customers to use. The company created AI Hub, which focuses on using large language models to process and understand information in documents without the need for users to fine-tune or train ML models.

Instabase is multicloud, but in September 2022, it chose to create its software-as-a-service (SaaS) stack fully on Amazon Web Services (AWS) for improved performance, availability, and cost. The company built its AI Hub solution on this SaaS stack. Using AWS for its new SaaS solution, Instabase saves on costs, improves its speed to market and time to value, and achieves a solution that is reliable and highly available.



Opportunity | Using Amazon EC2 to Build SaaS Solution AI Hub for Instabase

Instabase was founded in 2015 and specializes in processing semistructured and unstructured data of various formats in the intelligent document processing space. With the popularity of products launched into the GenAI market in November 2022, Instabase heavily invested in building its product using large language model technology. In February 2023, Instabase started building an AI Hub prototype with a simple user experience so that customers could upload their documents and ask questions about the data. With AI Hub, Instabase wanted to remove the complex and time-consuming process of training ML models, and that is where GenAI and the SaaS technology stack came in.

Instabase launched its SaaS stack in April 2023 and chose to migrate from multicloud to AWS for improved performance, availability, reliability, and costs. "Choosing AWS was the right thing to do and r



simpler,” says Shaunak Godbole, director of engineering at Instabase. “Whatever tools and architecture we can think of are already available from AWS, and we no longer have to worry about the complexities of the cloud.”

AI Hub is built entirely on AWS and runs on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. Instabase migrated to AWS for all its SaaS development and relied on support from AWS teams whenever it had questions. “The AWS team was very helpful in answering questions for us,” says Godbole. “The turnaround time was something I’d never seen before, from a few minutes to an hour, and we had all the support we needed.”

Solution | Saving 30 Percent on Costs While Improving Performance and Availability Using AWS

When a document is uploaded, AI Hub digitizes the document and encodes it before processing it through the selected large language model.

The whole system runs on Amazon EC2 Instances, and because there are appropriate instances for every workload, Instabase saves 30 percent on compute costs. For example, the solution runs on [Amazon EC2 M6i Instances](#), general-purpose instances powered by third Generation Intel Xeon Scalable processors. By migrating the solution to Amazon EC2 M6i Instances, the company saw an automatic 20 percent improvement in performance. Instabase also runs between 30 and 40 stacks on AWS for development purposes. Because development environments are created for a short time, the company uses [Amazon EC2 Spot Instances](#)—used to access spare Amazon EC2 capacity—for these stacks, and by doing so, it saves 70 percent on related costs.

Another important aspect of AI Hub was that it needed to be highly available. Instabase chose to use AWS services with that availability built in. Core to AI Hub is [Amazon Elastic Kubernetes Service](#) (Amazon EKS)—a managed Kubernetes service to run Kubernetes on AWS—on which Instabase has built the entire solution and where the orchestration happens. For its search system, Instabase uses [Amazon OpenSearch Service](#), made to securely unlock near-real-time search, monitoring, and analysis of business and operational data. The high availability of these services means Instabase can focus on innovation.

One key benefit of using AWS is reliability. Instabase uses [Amazon Simple Storage Service](#) (Amazon S3), object storage built to retrieve any amount of data from anywhere, for the block-storage of its file system. Amazon S3 has phenomenal reliability, which is important for Instabase because we are dependent on the service.



system,” says Godbole. The company also uses many of the available features of Amazon S3, including hot and cold storage.

Instabase improved time to value and speed to market using AWS. The company rearchitected in 3 months and launched a new product with some applications prebuilt. Deploying the SaaS solution was dramatically shortened, with efficiencies from provisioning to work for the infrastructure teams to delivering the product quickly for customers. Instabase can now stand up environments up to 95 percent faster than on-premises deployments. “The process was surprisingly smooth,” says Godbole. Using AWS also served Instabase in meeting compliance requirements such as SOC 2. The controls required for SOC 2 were already in place using the AWS service, and the company could meet compliance quickly.

Outcome | Innovating with New Technologies Using AWS

Its launch in June 2023 was only the beginning for AI Hub. There are new features and improvements in its product pipeline, and using AWS, Instabase doesn’t have to think about the infrastructure and can innovate more freely. The company is also using AWS to pivot its solutions to new technologies and improve things such as multitenancy in a more secure way.

“Support from the AWS team has been fantastic, not only for changing our old architecture, but also in implementing our road map for the future,” says Godbole.

About Instabase Inc.

Instabase is an AI application that transforms complex, unstructured data into actionable insights using GenAI. It helps users access any data that flows through their organizations in any format and document type—even handwriting.

AWS Services Used

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon EC2



Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides sizable compute capacity in the cloud. It is designed to make web-scale computing easier for developers.

[Learn more »](#)

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

[Learn more »](#)

Amazon EC2 M6i Instances

General purpose instances powered by 3rd Generation Intel Xeon Scalable processors.

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